TECHNICAL PROJECT DOCUMENTATION

CoroVista

Cardiovascular Risk Visualization & Vessel-Specific Stenosis Prediction System

A machine-learning workflow for cardiovascular risk assessment, vessel-level prediction, and interpretable model outputs.



PROJECT TEAM

Team Parallel Minds

TEAM MEMBERS

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DOCUMENTATION NOTE

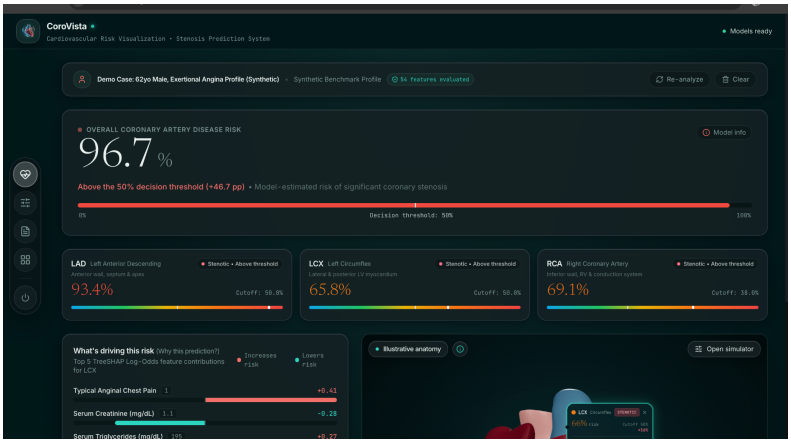
Prepared from repository documentation. Performance metrics are identified as repository-reported, not independently validated.

01 Project Overview

CoroVista combines multi-target CAD prediction, patient-level SHAP explanations, an interactive 3D coronary digital twin, and a 54-parameter patient simulator.

Scope note: Research and educational prototype only — not a validated diagnostic device.

Platform at a Glance



Core module	Function
Prediction	Cath + LAD / LCX / RCA
3D digital twin	Risk-coloured coronary branches
Simulator	54 editable clinical parameters
SHAP	Local feature attributions

Document Map

Sec.	Section	Coverage
02	**System architecture**	Application flow and 3D digital twin
03	**Data and preprocessing**	Dataset and input preparation
04	**Models and reported results**	Model metrics and thresholds
05	**Explainability and API**	SHAP, endpoints, setup and testing
07	**Limitations and roadmap**	Responsible use and future work

Prediction Models

Target	Model	Cutoff
Cath	XGBoost + Platt	0.50
LAD	XGBoost + Platt	0.50
LCX	XGBoost (uncalibrated)	0.50
RCA	Logistic Regression + Platt	0.38

Typical User Workflow

01 · INPUT	02 · PROCESS	03 · PREDICT	04 · EXPLORE
Enter patient data	**Validate and prepare**	**Review vessel scores**	**Inspect explanations**
Use the 54-parameter simulator.	The API aligns inputs for inference.	View Cath, LAD, LCX and RCA outputs.	Explore SHAP and the 3D risk overlay.

02 System Architecture

The diagram summarizes the request path: the frontend collects and presents information; the backend validates and routes requests; preprocessing aligns inputs with trained models; predictions and explainability outputs return to the interface.

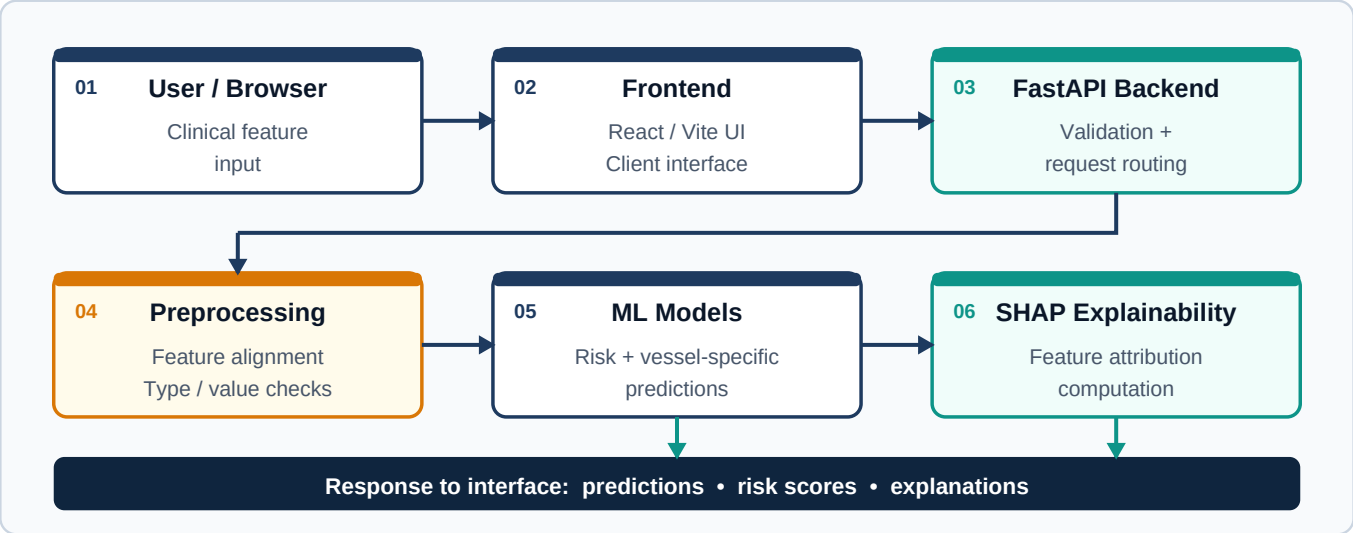
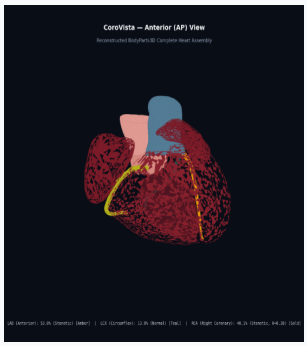


Figure 1. Conceptual application and inference flow based on repository structure and documented functionality.

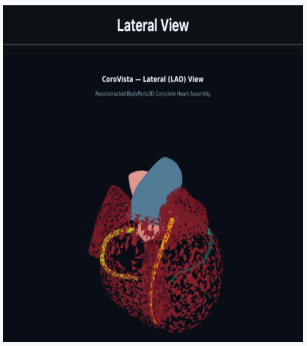
Interactive 3D Coronary Anatomy



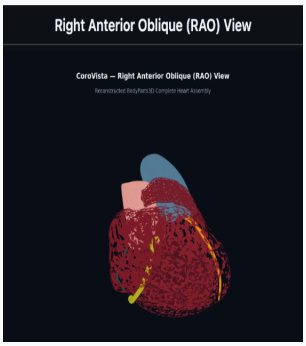
Anterior · LAD



Posterior · LCX



Lateral



RAO view

3D Digital-Twin Data Flow

Clinical inputs	Inference API	Vessel probabilities	3D risk overlay
54 parameters / patient archetype	Validation, preprocessing and four target models	Cath, LAD, LCX and RCA scores	Risk-coloured coronary branches in the interactive heart viewer

The 3D colour map visualizes predicted vessel-level risk; it does not localize a physical lesion or represent plaque geometry.

How to Read the Architecture

01 · Input layer
The user enters or selects a 54-feature patient profile.

02 · Inference layer
The API validates inputs and returns target-specific model scores.

03 · Visualization layer
The interface presents vessel scores and explanatory outputs.

03 Data and Preprocessing

The repository identifies the **Extended Z-Alizadeh Sani** dataset as the principal dataset (**303 records**, **54 features**) and includes a preprocessing pipeline to transform incoming records into model-ready inputs.

Preprocessing Considerations

- **Feature alignment:** keep inference columns consistent with the features used during training.
- **Data types:** represent numerical and categorical/string columns in formats accepted by the pipeline.
- **Missing or malformed values:** validate requests and handle unsupported values rather than silently treating them as valid observations.
- **Inference consistency:** apply the same transformations at prediction time as during training.

Data Summary

Item	Documented Value	Meaning
Dataset	Extended Z-Alizadeh Sani	Dataset named in project documentation
Records	303	Reported dataset size
Features	54	Reported feature count

04 Models and Reported Results

The repository reports **ROC-AUC** (ranking discrimination across thresholds) and **PR-AUC** (precision-recall trade-off under class imbalance).

Repository-Reported Metrics & Thresholds

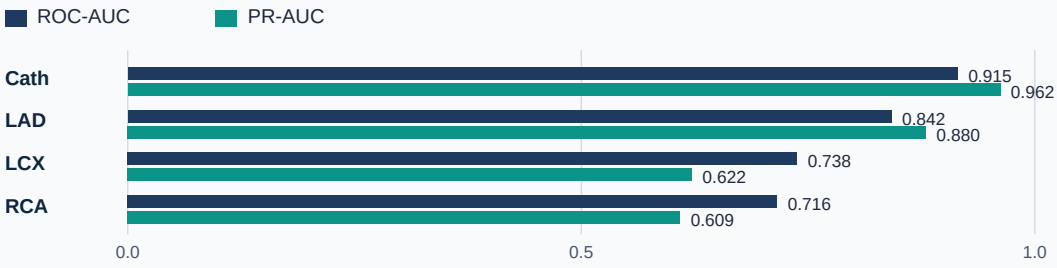
Target / Model	ROC-AUC	PR-AUC	Threshold
Cath — XGBoost	0.915 ± 0.041	0.962 ± 0.020	0.50
LAD — XGBoost	0.842 ± 0.048	0.880 ± 0.042	0.50
LCX — XGBoost	0.738 ± 0.061	0.622 ± 0.087	0.50
RCA — Logistic Regression	0.716 ± 0.057	0.609 ± 0.073	0.38

Thresholds recorded in project documentation: **Cath 0.50**; **LAD 0.50**; **LCX 0.50**; **RCA 0.38**.

Interpretation Notes

- The **±** values are reproduced as documented; their statistical meaning should be confirmed from experiment artifacts.
- These are repository-reported results, not independently reproduced benchmarks; evaluation splits should be checked in experiment records.
- A high AUC alone does not establish clinical usefulness, calibration, safety, or generalization.

AUC Comparison (Repository-Reported)



05 Explainability and API Surface

CoroVista includes SHAP-based feature attribution to estimate how input features contribute to a model output relative to a reference distribution (describing model behavior, not causation).

Endpoint	Purpose
`/api/v1/health`	Health/status check; may report whether models are loaded.
`/api/v1/models`	Model metadata or available model information.
`/api/v1/features`	Feature metadata used by the application.
`/api/v1/predict`	Prediction request endpoint.
`/api/v1/explain`	Explanation/feature-attribution request endpoint.
`/api/v1/analyze`	Combined analysis flow for predictions and explanation information.

Typical Request Lifecycle

- Client submits a structured feature payload.
- Backend validates the request and runs preprocessing.
- Selected model generates predictions; explainability path computes attributions.
- Backend returns a structured response for the frontend to render.

Confirm exact schemas against the running OpenAPI page (/docs).

06 Local Setup and Testing

Local development used a Python virtual environment, FastAPI/Uvicorn backend, and Vite frontend. The backend uses port **8001** because another service occupied port 8000.

```
# Backend (FastAPI / Uvicorn on port 8001)
cd ~/Projects/CoroVista && source .venv/bin/activate
uvicorn backend.app.main:app --host 127.0.0.1 --port 8001 --reload

# Frontend (React / Vite)
cd frontend && npm run dev
```

Expected URLs: frontend http://localhost:5173; backend http://127.0.0.1:8001; API docs http://127.0.0.1:8001/docs.

Verification Checklist & Testing Notes

- Confirm the health endpoint responds and reports expected model-loading status.
- Ensure frontend API base URL points to port 8001 and CORS allows localhost:5173 / 127.0.0.1:5173.
- Run an analysis request and inspect predictions and explanation payloads.
- Check logs for model-loading errors, feature mismatches, or dependency incompatibilities.

Testing notes: Earlier local integration notes recorded a successful health check and synthetic analysis response with **HTTP 200** (including prediction and SHAP outputs). The repository README currently reports **62 Pytest passed** and **50 Vitest passed**; these are repository-reported counts and should be rerun against the final commit before making release-readiness claims.

3D Coronary Risk Colormap

Risk band	Probability range	Visual meaning
Low	$p < 30\%$	Emerald / green
Moderate	$30\% \le p < 50\%$	Amber / yellow
High	$50\% \le p < 75\%$	Orange
Critical	$p \ge 75\%$	Rose / red

Risk bands are visualization thresholds documented in the README, not clinical diagnostic categories.

07 Limitations, Responsible Use and References

Key Limitations

- **Clinical validation:** performance on a project dataset does not establish safety for clinical decisions.
- **Generalization:** results may change across institutions, populations, measurement protocols, and prevalence levels.
- **Calibration:** discrimination metrics alone do not show whether predicted probabilities match observed event rates.
- **Explainability:** SHAP is not causal and can depend on background data and feature dependence.
- **Security and privacy:** real-world clinical data handling requires access controls, secure transport, retention rules, and privacy review.

Responsible-Use Statement

CoroVista should be presented as a research and educational prototype. Outputs should be reviewed by qualified clinicians and must not be used as the sole basis for diagnosis, treatment, or patient management. Real-world deployment would require external validation, calibration assessment, clinical workflow evaluation, privacy safeguards, and regulatory review.

References and Source Notes

- CoroVista project repository: https://github.com/Nikhil9897/CoroVista
- Dataset name and metrics are reproduced from project documentation and experiment summaries.
- SHAP documentation: https://shap.readthedocs.io/

Future Improvements & Roadmap

The following items are listed as future improvements in the repository README; they are proposed roadmap work, not claims of completed functionality.

Workstream	Planned improvement	Expected benefit
Deployment	Add Docker and Docker Compose definitions.	Repeatable one-command multi-service setup.
Clinical workflow	Explore HL7 / FHIR adapters for standard EHR summaries.	Reduce manual entry when integrating supported clinical data.
Spatial UI	Improve layouts for simultaneous 3D manipulation and parameter tuning.	Make simulator and anatomy views easier to use together.
Performance	Apply dynamic code-splitting to Three.js / React Three Fiber bundles.	Reduce initial frontend bundle loading cost.

Roadmap at a Glance

01 · Deploy	02 · Integrate	03 · Improve UI	04 · Optimize
Docker / Compose	HL7 / FHIR adapters	3D + simulator layout	Bundle code-splitting

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